This preprint has been published in *European Journal of Personality* (https://doi.org/10.1177/08902070241251516). This paper is not the copy of record and may not exactly replicate the authoritative version of the article.

How much can personality predict prosocial behavior?

Yngwie Asbjørn Nielsen[1,2], Stefan Pfattheicher[1], & Isabel Thielmann[2]

[1]Aarhus University

[2]Max Planck Institute for the Study of Crime, Security and Law, Freiburg

Word count: 9,139

Display items: 4

Corresponding author

Correspondence regarding this manuscript should be addressed to Yngwie Asbjørn Nielsen, Aarhus University, Jens Chr. Skous Vej 2, 8000 Aarhus C, Denmark. E-mail: yan@cc.au.dk

Funding

The authors received no funding in support for this research.

Competing interests

The authors declare no competing interests.

Author contributions

Yngwie Asbjørn Nielsen: Conceptualization, Methodology, Formal Analysis, Writing – Original Draft, Writing – Review & Editing. Stefan Pfattheicher: Conceptualization, Writing – Review & Editing. Isabel Thielmann: Conceptualization, Data Curation, Writing – Review & Editing.

Abstract

Explaining prosocial behavior is a central goal in classic and contemporary behavioral science. Here, for the first time, we apply modern machine learning techniques to uncover the full predictive potential that personality traits have for prosocial behavior. We utilize a large-scale dataset (N = 2,707; 81 personality traits) and state-of-the-art statistical models to predict an incentivized measure of prosocial behavior, Social Value Orientation (SVO). We conclude: (1) traits explain 13.9% of the variance in SVO; (2) linear models are sufficient to obtain good prediction; (3) trait–trait interactions do not improve prediction; (4) narrow traits improve prediction beyond basic personality (i.e., the HEXACO); (5) there is a moderate association between the univariate predictive power of a trait and its multivariate predictive power, suggesting that univariate estimates (e.g., Pearson's correlation) can serve as a useful proxy for multivariate variable importance. We propose that the limited usefulness of nonlinear models may stem from current measurement practices in personality science, which tend to favor linearly related constructs. Overall, our study provides a benchmark for how well personality predicts SVO and charts a course toward better prediction of prosocial behavior.

Keywords: Machine learning, Benchmark, Personality, Prosocial behavior, Social Value Orientation (SVO)

Word count: 186

Plain language summary

The importance of being kind, helpful, and cooperative—of behaving *prosocially*—is undeniable. And yet, not everyone is equally prosocial. In fact, research in the social and behavioral sciences consistently shows that prosociality differs between individuals and is predicted by individuals' personalities. This research, however, typically only manages to explain a small portion of the variance in prosocial behavioral tendencies, rarely going beyond 9% (corresponding to a correlation of $r = .30$). One possible explanation for the lack of better prediction is the widespread use of constrained statistical models including only few personality traits as predictors at a time. Using a large set of predictors and state-of-the-art statistical models, we estimate that personality traits account for 13.9% of the variance in Social Value Orientation, an established indicator of prosocial behavior. By modeling multiple personality traits in tandem, we thereby break the 9% ceiling—a feat that single-predictor models rarely manage. However, we also find that more complex (i.e., nonlinear) statistical models only slightly increase the amount of variance explained compared to more restrictive (i.e., linear) models. This suggests that the pursuit of better personality models of prosociality does not hinge on increasing the complexity of statistical models; instead, we suggest it starts by embracing a wider range and variety of personality traits.

Word count: 212

Introduction

Prosocial behaviors (i.e., acts that benefit others) represent perhaps the brightest side of human nature. Individuals donate organs to strangers, nurture the young and the old, build tightly knit communities, and sacrifice their own wellbeing to protect the climate for future generations (Thielmann & Pfattheicher, 2022). However, not everyone is equally prosocial. Previous research has documented extensive individual variation in the propensity to behave in prosocial ways (e.g., Fehr & Fischbacher, 2002; Hilbig et al., 2014; Murphy & Ackermann, 2014), spawning an impressive research endeavor into the personality basis of prosocial behavior (Thielmann et al., 2020). At the heart of this endeavor lies the goal of mapping personality traits onto prosocial behavior—or, stated differently, to *predict* prosocial behavior from personality. Critically, however, most personality predictors are limited in their explanatory power, accounting for less than 5% of the variance in prosocial behavior (Thielmann et al., 2020). In the present investigation, we explore whether applying machine learning techniques that account for potential nonlinear relations and make use of a large set of personality predictors can improve the predictive power of personality to answer the question "how much can personality predict prosocial behavior?". We do so by drawing on the specific example of Social Value Orientation (SVO) as an established indicator of prosocial behavioral tendencies (Balliet et al., 2009; McClintock & Liebrand, 1988; Murphy & Ackermann, 2014).

Beyond univariate prediction

Although prediction is a major goal in many scientific fields (Yarkoni & Westfall, 2017), the search for strong predictors plays a particularly prominent role in the history of the social sciences, especially in personality research (Mõttus et al., 2020). During the decades-old person-situation debate (Epstein & O'Brien, 1985; Pervin, 1989), personality research was heavily criticized for only accounting for (very) small portions of variance in certain outcomes, at best reaching 9% (or, equivalently, $r = .30$; Mischel, 1968). Consequently, some have argued that the study of personality traits should be abandoned altogether (Shweder, 1975). The debate was eventually resolved (see, e.g., Fleeson, 2004; Hogan, 2009; Kenrick & Funder, 1988), and personality science persevered—indeed, the use of personality measures spread to many fields within and beyond psychology, including sociology, behavioral economics, and medicine (e.g., Chapman et al., 2011; Turner, 1988). Even so, most effect sizes in personality research remain modest to this day (i.e., $r < .30$; Bosco et al., 2015; Funder & Ozer, 2019; Richard et al., 2003), and the case of personality and prosocial behavior is no different (i.e., the strongest predictor identified in the meta-analysis by Thielmann et al., 2020, attained $\hat{\rho} = .26$).

The apparent stability of this finding suggests that the predictive power of personality psychology hit a ceiling fifty years ago and never advanced beyond it.[1] But is this truly the upper bound of our ability to predict prosocial behavior via personality? Perhaps not. Most of the field has focused on univariate prediction, that is, using a single trait to predict behavior (Thielmann et al., 2020). Modern predictive models, by contrast, can leverage the unique contributions of hundreds (or thousands) of predictors at once. The disregard for multivariate prediction may have been limiting the predictive capabilities of the field. Indeed, a simulation study by Ahadi and Diener (1989) showed that if a behavior is determined by multiple traits, a single trait obtains an upper bound on its predictive capacity (e.g., a behavior determined by four traits has a limit of $r = .45$ for a single trait). Thus, a modern look at predicting behavior from personality demands going beyond univariate models.

Benchmarking personality and prosocial behavior

The question of how well personality predicts prosocial behavior is perhaps more pertinent than ever. Recently, multiple scholars have argued that prediction ought to be an important scientific goal on par with description and explanation (e.g., Mõttus et al., 2020; Yarkoni & Westfall, 2017). At the same time, the adoption of machine learning techniques has provided social and behavioral scientists with the tools to work rigorously toward better predictive models (Bleidorn & Hopwood, 2019; Rosenbusch et al., 2021). In particular, the practice of *benchmarking*—that is, evaluating statistical models based on their predictive performance on a well-defined task (typically an openly available dataset) and publishing the result for others to compare with or beat—has been highlighted as a central tool in securing a cumulative scientific endeavor (Rocca & Yarkoni, 2021).

We set out to provide the first benchmark of the personality–prosociality relation, essentially testing how much of the variance in prosocial behavior can be explained by personality *in general*. Answering such a question requires the use of machine learning techniques that are tailored to optimally make use of large datasets and account for the potentially complex relations between variables (Rosenbusch et al., 2021). If, for example, personality traits interact or are nonlinearly related to prosocial behavior, then a traditional linear model may underestimate the strength of the relation. However, adopting machine learning techniques substantially widens the space of choices the analyst must make. Thus, in addition to providing a benchmark of

[1] We note that two traits may (and often do) correlate above $r = .30$, especially if they share considerable conceptual overlap (Bainbridge et al., 2022). Nonetheless, the relation between personality traits and *behavioral* measures of prosociality remains at a small-to-medium size.

the field's ability to predict prosocial behavior, we chart a course toward better predictive models.

The present research: A case study on Social Value Orientation

There exists a plethora of measures of prosocial behavior, but for the present investigation we will focus on a single, widely used measure, namely SVO (Murphy et al., 2011).[2] An individual's SVO describes a relatively stable pattern in how they weigh the importance of their own outcome versus other people's outcomes and is typically measured using a series of resource-allocation decisions (Balliet et al., 2009; McClintock & Liebrand, 1988; Murphy & Ackermann, 2014). SVO shows strong ties to other measures of prosocial behavior, including economic game-like paradigms (Thielmann et al., 2020) as well as naturalistic, real-life measures (e.g., Cameron et al., 1998; Van Lange et al., 2007, 2011). Since SVO is measured using multiple decisions, it reflects individual differences more reliably than single-shot economic games (De Matos Fernandes et al., 2022; Murphy et al., 2011). Moreover, the SVO measure can be incentivized with actual money, rendering the decisions consequential. For these reasons, we take SVO as our outcome.

In a preregistered analysis, we apply 15 machine learning techniques to predict SVO from 81 personality predictors measured in the Prosocial Personality Project (PPP), a large-scale project covering various personality traits and social behaviors. Specifically, we fitted a variety of linear models that attempt to select only the best personality predictors, two such linear models that included pairwise interactions between personality predictors, and four powerful nonlinear models. We thereby go beyond prior research, which has primarily dealt with univariate prediction and heavily constrained (i.e., linear) models, and provide a thorough benchmark of the relation between personality and SVO.

Methods

Sample

The Prosocial Personality Project. Our analyses rely on data from the PPP. The measures included in the PPP were chosen following an exhaustive meta-analysis summarizing six decades of research on personality and prosocial behavior in economic games (Thielmann et al., 2020), which was then extended by a thorough literature search for constructs related to prosociality and related constructs (for more details, see https://osf.io/m2abp/). Since the PPP

[2] SVO is often conceptualized as a disposition (e.g., Murphy & Ackermann, 2014). However, the measure entails a series of choices and is, thus, also commonly used as a behavior (e.g., Hilbig et al., 2023).

arguably covers the breadth of known prosocial personality traits, we permit ourselves to make claims concerning this field of research as a whole and its present capacity for predicting SVO.

The PPP data were collected online over multiple measurement occasions between November 2019 and May 2023 via a panel provider in Germany. For the current investigation, we used data from waves T1–T5 as well as follow-up waves 2020-05b and 2021-10. Pre-specified exclusion criteria slightly varied from wave to wave, but generally included failing instructed attention checks, extremely low variance in responses, low response times, inconsistencies in demographic variables across waves, and multiple participation in one and the same wave. Detailed information on sampling, exclusion criteria, payment, etc. is available on the Open Science Framework (OSF; https://osf.io/m2abp/). Data from the PPP has been used in prior publications, of which the full list is also available on the OSF (https://osf.io/8xhkd). Of these, one prior publication concerned SVO (Hilbig et al., 2023), but none have investigated the research question addressed here. Notably, Hilbig et al. (2023) only looked at univariate prediction.

Participants. Participants for the PPP were recruited via a German panel provider with the goal of obtaining a sample that is approximately representative of the German population with regard to age, gender, and educational level. The present investigation draws on all participants in the PPP which responded to wave T5 containing the SVO measure. The sample comprised 2,707 participants (1,286 identified as female, 1,419 as male, 2 had another gender identity) between 18 and 78 years (M = 43.2, SD = 12.5) with diverse educational backgrounds (942 held a degree from university, 766 from high school [German: *Abitur*], 967 from secondary school [German: *Realschulabschluss*], 28 from secondary general school [German: *Hauptschulabschluss*], 1 none, 3 other) and employment statuses (1709 were employed, 223 independent, 85 unemployed, 234 retired, 115 homemaker, 273 student or apprenticeships, 68 other).

Measures

Social Value Orientation. SVO was measured using the primary items from the SVO Slider Measure (Murphy et al., 2011). Participants make six decisions on how to allocate money between themselves and another person, each time choosing between nine options (e.g., "You receive 85; Other receives: 68"). SVO is calculated as the angle between the mean payoff allocated to the other person (denoted $\bar{A}_o$) and to the participant themselves ($\bar{A}_s$; Murphy et al., 2011):

$$\text{SVO°} = \arctan\left(\frac{\bar{A}_o - 50}{\bar{A}_s - 50}\right)$$

To incentivize the decisions, one of the six decisions was randomly chosen, and the corresponding distribution of money as selected by the participant was paid out to the participant and a randomly assigned other individual as a bonus payment, using a conversion rate of €0.02 for each point earned (M = €1.55, SD = €0.30, range: €0.30 to €2.00).

Predictors. The project included 81 trait predictors in the realm of basic personality, prosociality, self-control, morality, and psychopathology (e.g., the HEXACO, Big Five Agreeableness, PID-5, Dark Tetrad). The full list of traits is shown in Table 1.

[Table 1 goes here]

Analytic strategy

A brief introduction to nonlinear models. Nonlinear models allow researchers to uncover complex statistical relations in data that go beyond the linear constraint imposed by conventional statistical analyses. There exist a wide variety of nonlinear models, some of which work very differently from the linear model; for example, a *random forest* works by fitting thousands of decision trees (e.g., "if X_1 is above 2.0 and X_2 is below 1.5, predict that y equals 3.2") on bootstrap samples of the data and averaging their predictions (see James et al., 2013, for an introduction to many of the models considered here). One challenge that arises with such complex models is that they can pick up arbitrary noise in the dataset and thereby dramatically inflate the apparent accuracy of a model (known as *overfitting*; Rocca & Yarkoni, 2021). To counter this, one must evaluate models on different data than it is fitted to—a procedure known as *cross-validation*. In our case, we fitted the models on 70% of the dataset, and tested their predictions on the remaining 30%. An additional challenge is that many machine learning models contain so-called *hyperparamters* (or *tuning parameters*) which constrain or control the way they are fitted. For example, the random forest contains one hyperparameter specifying the number of decision trees, and another controlling the maximum number of branches per tree. The optimal choice of hyperparameters is unknown and must be determined by the analyst (Probst et al., 2019). To do so, one common approach is to perform a *grid search*, whereby many models are fitted across a range of hyperparameter values, and the best performing model is selected using cross-validation (Bischl et al., 2023).

Statistical models. We ran a plurality of statistical models. These can be grouped into three types: (1) linear additive models, (2) linear models including pairwise interaction terms

obeying a strong hierarchy constraint, that is, where inclusion of an interaction effect implies inclusion of the main effects (Bien et al., 2013), and (3) nonlinear models. The full list of models and their respective hyperparameters is shown in Table 2.

[Table 2 goes here]

Analytic strategy. First, the dataset was split 70:30 into a training set (n = 1,894) and a test set (n = 813) and all predictors were standardized based on the training set. The linear models and models with interactions were tuned with a grid search. The deep neural network was tuned using early stopping and a validation split of .20. All other nonlinear models were tuned using Bayesian model-based optimization running for 1,000 iterations (half as initial exploration). The optimal set of hyperparameters was identified using the 10-fold cross-validation MSE. After training, the models were evaluated on the test set. Specifically, we computed the R^2, defined as 1 minus the ratio between the variance of the residuals and the total variance in the test data. To isolate the predictive power of a basic personality model (i.e., the HEXACO, the entire procedure was then repeated using only the HEXACO dimensions as predictors.[3] All models and results from tuning are available on the OSF (https://osf.io/zcdk8).

Variable importance. To quantify the relative importance of each trait for predicting SVO, we calculated permutation-based variable importance (Fisher et al., 2018). First, a trait variable is permuted, that is, the observations of the variable are shuffled around, thereby effectively turning it into noise. Next, a previously fitted model is used to predict SVO from the permuted variable. Variable importance is then calculated by dividing the permuted model's mean squared error (MSE) with the original (non-permuted) model's MSE. If the variable was important, the predictions from the permuted model should worsen (i.e., the MSE should increase), in turn yielding an importance score above 1. This procedure is repeated five times for each predictor and for each model.[4]

Missing data. We included any participant having complete data on the SVO Slider in the analysis. Missing data (15% in total, mostly present in seven predictor variables coming from the same wave, i.e., follow-up 2021-10) was imputed via random forests. To estimate the level of uncertainty due to missingness, we imputed m = 5 datasets. Imputation was done

[3] The HEXACO model extends the Big Five by a sixth basic trait dimension, titled Honesty–Humility, that captures proactive prosociality (see Ashton et al., 2014). For a detailed discussion regarding other differences between the two personality models (i.e., between Emotionality and Neuroticism, and between Agreeableness in the two models), see Thielmann, Moshagen, et al. (2022).

[4] We relied on the R package iml (Molnar & Schratz, 2022) to compute variable importance. As such, we were unable to investigate variable importance in the deep neural network, which was fitted using python.

separately on the train and test sets. The analysis was then conducted independently on each imputed dataset and the final estimates were pooled by averaging. Imputation diagnostics (i.e., trace lines and densities) are available on the OSF (https://osf.io/zcdk8).

Software. Most analyses were performed in R (R Core Team, 2021) and RStudio (RStudio Team, 2020). Missing data were imputed with mice (Buuren & Groothuis-Oudshoorn, 2011). Forward-backward selection was tuned with caret (Kuhn et al., 2022), relying on leaps (Miller, 2020). The lasso, ridge, relaxed lasso, and adaptive lasso regressions were fit using glmnet (Friedman et al., 2021). The elastic net was tuned with ensr (DeWitt & Bennett, 2019), also relying on glmnet. The SCAD and MCP models were fitted with ncvreg (Breheny, 2021), and the non-negative garrote with nnGarrote (Christidis et al., 2021). The models with interactions were fit with hierNet (Bien & Tibshirani, 2020) and glinternet (Lim & Hastie, 2015), respectively. Most nonlinear models were tuned using mlr3 (Lang et al., 2022), relying on xgboost (Chen et al., 2022), ranger for the random forest (Wright et al., 2022), and kernlab for the support vector regression (Karatzoglou et al., 2022). Variable importance measures were derived using iml (Molnar & Schratz, 2022), and the spline regression was fitted with segmented (Muggeo, 2023). The neural network was modelled using Python (Van Rossum & Drake, 2009), relying on keras (Chollet et al., 2015) for fitting and hyperopt (Bergstra et al., 2015) for tuning.

Deviations from preregistration

The analysis deviated from the preregistration in the following ways. Three deviations were made to improve the quality of the modeling: (1) the data were split 70:30, not 50:50, to make better use of the data for training, (2) the nonlinear models were tuned via model-based optimization instead of grid search to improve the chances of finding a global optimum within reasonable time, and (3) the support vector regression was additionally tuned for type of kernel (radial, polynomial) and polynomial degree. Finally, one deviation was made to reduce the model runtime to a manageable level: (4) the deep neural network was tuned using a validation split and early stopping instead of 10-fold cross-validation. The analyses of variable importance were not preregistered.

Results

Personality traits can explain 13.9% of variance in SVO

Looking at Figure 1, the best-performing model (i.e., the deep neural network [DNN]) explained 13.9% of the variance in SVO. In other words, our results suggest that one seventh of SVO is attributable to personality traits. By contrast, the strongest univariate predictor, competitive jungle word view, explained 9.3% of the variance (see Table 1). This corresponds to a correlation of r = .31 and thus coincides with the r = .30 ceiling on prediction popularized by Mischel (1968). This comparison reveals that a substantial increase in predictive performance can be achieved by modeling multiple traits in tandem; indeed, multivariate models are able to clearly break the 9% ceiling.

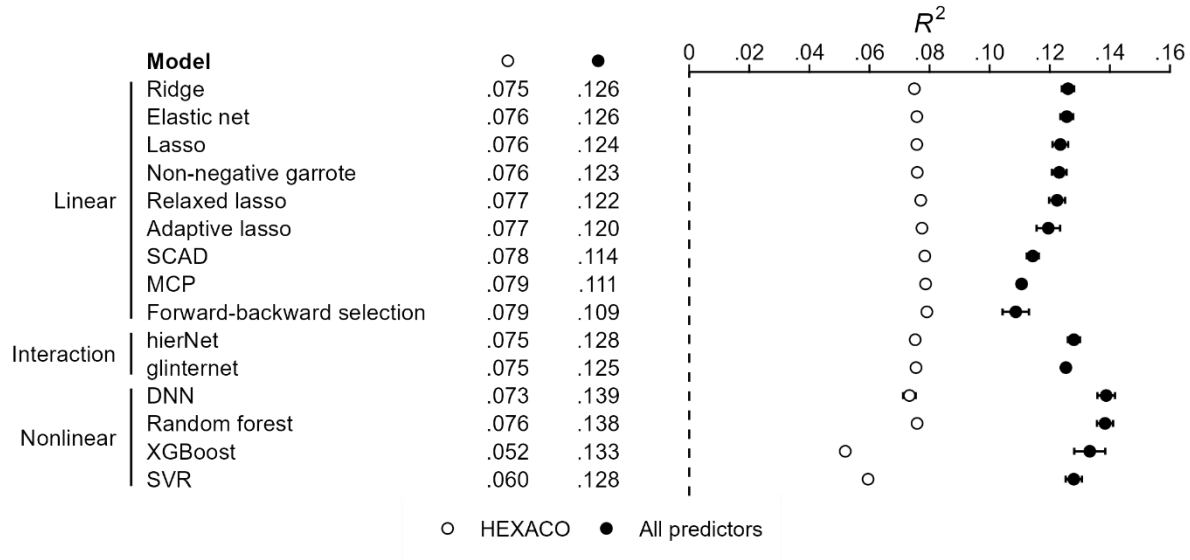


Figure 1. Cross-validated R^2 for predictive models of SVO

Cross-validated R^2 for a selection of linear models (forward-backward selection, lasso, ridge, elastic net, relaxed lasso, adaptive lasso, SCAD, MCP, non-negative garrote); linear models with pairwise interactions obeying strong hierarchy (hierNet, glinternet); and nonlinear models (XGBoost, random forest, deep neural network [DNN], support vector regression [SVR]). Error bars denote the standard error across imputed datasets and may be evidence of uncertainty due to missingness or randomness inherent to the model fitting procedure.

Although this analysis provides a snapshot of the present predictive capabilities of the field, it does not by itself tell us which direction the field should move towards to improve upon those predictions. For example, it may be that using more complex (e.g., nonlinear) models could better leverage the predictive potential of personality measures for prosocial behavior. Alternatively, the answer may lie in developing better predictors. We turn to this matter next.

Linear models are sufficient

Linear models (e.g., Pearson's correlation) make extremely restrictive assumptions about the relation between variables, thereby potentially failing to adequately capture and leverage the (possibly nonlinear) pattern of the data. Even so, in our analyses, the best-performing linear model was able to attain an R^2 of 12.6%—only about one percentage point less than the best nonlinear model. This suggests that the effect of each predictor is mostly linear and thus well-modelled by traditional means (see Oparina et al., 2022, for a similar conclusion in the context of wellbeing). Although these results remain to be generalized to measures beyond SVO, they suggest that researchers in the field of personality and prosocial behavior may rest assured that the currently dominant practice of using linear models is sufficient.

Interactions are *not* useful

Theoretically, there has been an interest in estimating trait–trait interactions (see Bauer & Shanahan, 2007; Vize et al., 2022). For instance, the "effect" of a given trait may be more pronounced in individuals having a particular level on another trait (i.e., a moderation effect). We tested the utility of modeling interactions between personality traits using two models that attempt to select the best subset of predictors and their pairwise interactions. The best performing model with interactions explained 12.8% of the variance (Figure 1), showing that including interactions led to a miniscule improvement in prediction beyond the linear additive model, visible only at the third decimal. Moreover, the models including interactions introduced 41 and 66 interaction terms, respectively, thereby dramatically inflating the size of the model.[5] Taken together, modeling interactions between personality traits does not appear to be a fruitful approach for predicting SVO (see Vize et al., 2022, for a similar conclusion concerning the Big Five traits and a large set of life outcomes).[6]

[5] The number of included interaction terms varied depending on the imputation (glinternet: 36–66, hierNet: 44–97). The reported numbers are medians across imputations.

[6] We refrained from considering higher-order interactions (i.e., three-way interactions, four-way interactions, etc.) in our analysis due to their limited use in the field. Although it is possible that such interactions could potentially increase the predictive capability of the models, higher-order interactions are unlikely to be strong if the lower-order interactions are not (a phenomenon known as the *sparsity principle*; Li et al., 2006).

Narrow traits improve prediction

Figure 1 compares the predictive utility of a model trained only on the HEXACO personality model to the models trained on the entire predictor set (containing basic as well as narrow traits). The HEXACO is a basic personality model, developed to represent the entire personality space via six broad dimensions (Ashton & Lee, 2007). Notably, one of the six dimensions, Honesty–Humility, has previously been found to be a strong predictor of prosocial behavior in economic games (Thielmann et al., 2020). As shown, however, the models restricted to the HEXACO dimensions consistently performed about five percentage points worse than models leveraging the entire predictor set (Figure 1), suggesting that modeling multiple narrow traits in tandem yields incremental validity beyond the personality space spanned by the HEXACO dimensions.

Taken together, our results suggest that increasing model complexity (in terms of nonlinearities and interactions) matters less for good predictions than obtaining a comprehensive set of predictors (i.e., going beyond basic personality). In other words, the best way to attain better prediction is, at present, a matter of developing and discovering new and better traits (and measures thereof), rather than increasing the modeling complexity of the existing traits (see also Oparina et al., 2022). Therefore, we finally present two findings that may be useful in the search for better predictors.

Pearson's correlations are informative

Most work in the social sciences has focused on estimating linear univariate relations (Funder & Ozer, 2019; Thielmann et al., 2020), as often captured in a Pearson's correlation coefficient. This approach, however, may misrepresent the true importance of a predictor if the underlying reality is nonlinear and/or multivariate. Fortunately, permutation-based variable importance accounts for non-straightforward ways in which a predictor may influence prediction, such as via nonlinearities, interactions, or suppressor effects[7]—all of which are neglected by Pearson's correlation. Even so, Pearson's correlation may be an adequate measure of variable importance—especially if the relations at play are mostly linear, as implied by Figure 1. Indeed, as shown in Figure 2 for three exemplary models, the univariate R^2 value of each of the 81 personality predictors (roughly speaking, the square of Pearson's correlation) was moderately related to permutation-based importance measures derived from each model. In fact, when

[7] Suppressor effects arise when the inclusion of a covariate increases the overall model performance by removing outcome-irrelevant variance in another predictor. A variable can be an important suppressor without being correlated with the outcome (Pandey & Elliott, 2010).

relating the two measures of variable importance using a spline regression (a type of linear regression where the regression line breaks at a "knot"), the univariate R^2 of the correlations explains between 19.0% and 63.4% of the variance in permutation-based importance, depending on the model. This shows that Pearson's correlation coefficient can approximate how important a personality trait is for predicting SVO, even when complexities are taken into consideration.

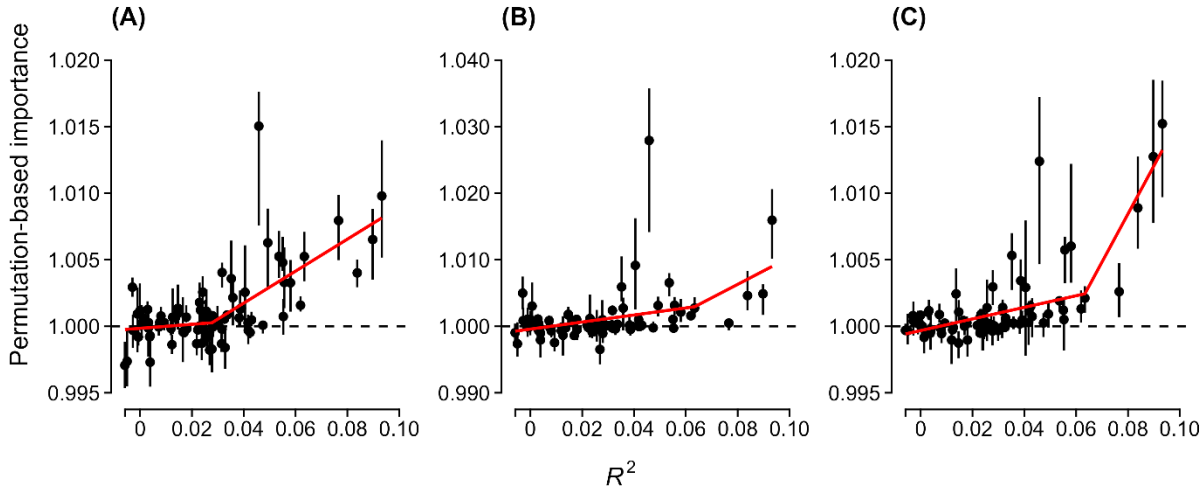


Figure 2. Univariate prediction is a proxy for multivariate variable importance

Relation between the strength of the univariate relation (expressed in R^2) of a predictor with prosocial behavior and its pooled importance estimates across a variety of multivariate models (where a value of 1 denotes no importance, and higher values denote higher importance). Each dot represents a personality predictor. A spline regression with a single knot is fitted to predict the importance of a predictor from its zero-order relation with SVO, as measured with a cross-validated R^2. Panel A contains importance values from a ridge regression. Knot position: R^2 = .028 (corresponding to r = .166); R^2 explains 45.4% of the variance in variable importance. Panel B contains importance values from a hierNet model. Knot position: R^2 = .063 (corresponding to r = .252); R^2 explains 19.0% of the variance in variable importance. Panel C contains importance values from a random forest. Knot position: R^2 = .063 (corresponding to r = .252); R^2 explains 63.4% of the variance in variable importance. Results from the remaining models are reported on the OSF (https://osf.io/zcdk8).

Correlations of at least .20 are important

Figure 2 reveals another interesting pattern; predictors that explain less than a certain proportion of variance in SVO on their own are of almost no importance in the models. That is, the line from the spline regression is almost completely horizontal before the "knot" value is reached. This pattern emerges in all models, although the exact position of the knot varies. On average, the knot is located at R^2 = .051, suggesting that "useful" predictors tend to (univariately) explain at least 5.1% of the variance in SVO, corresponding to a Pearson's correlation of r = .23.[8] Thus, it seems that for a predictor to be important *in the overall picture* it must as a rule-of-thumb attain a Pearson's correlation of at least r = .20. Curiously, this result coincides with the mean effect size in 20th century social and personality psychology (r = .21; Richard et al., 2003; see also Gignac & Szodorai, 2016). We suggest that this value also serves as a benchmark for researchers interested in the relation between personality and SVO. For instance, it may be used for power analyses when no better alternative to estimate the effect size is present (Lakens, 2022), and for evaluating the relative importance of the effect size of single predictors.

Discussion

Prosocial behavior is multiply determined by personality, situational factors, and their interaction (Furr & Funder, 2021; Li, 2023; Lewin, 1936). Given a specific situation (e.g., allocating money between oneself and a stranger), there are reliable individual differences that underlie prosociality (Thielmann et al., 2020). Continuing a long tradition in the social and behavioral sciences, we sought to thoroughly leverage the predictive potential of personality for a well-established measure of prosocial behavior, namely SVO. Using modern machine learning techniques that account for multivariate and nonlinear relations, we explained 13.9% of the variance in SVO by personality traits, thus establishing a benchmark for the prediction of prosocial behavior based on stable dispositions. This multivariate benchmark breaks the well-known 9% (or r = .30) ceiling on univariate trait models of behavior. In addition, we found that the hitherto dominant practice of using linear models of trait–outcome relations (without interactions) is sufficient to achieve good prediction.

[8] This was the only analysis we did not preregister. To ensure the result was not contingent on one specific analytic approach (see Multiverse analysis; Steegen et al., 2016), we applied multiple different models to validate the result. Specifically, we fitted a conditional inference tree, a spline regression, and a logistic regression. The resulting splits, knots, and inflection points were on average 0.044, 0.051, and 0.047, respectively (corresponding to a correlation coefficient between .21 and .23; for details, see the OSF: https://osf.io/zcdk8). The result thus appears robust to the type of analysis.

Rethinking benchmarks for personality and prosocial behavior

A crucial benefit of setting a benchmark is that it can serve as a reference point for what constitutes a good predictive model. Cohen (1988) provided three benchmarks for R^2, namely 2% (small), 13% (medium), and 26% (large). In light of our 13.9% benchmark, these values may not be applicable to the field of personality and prosocial behavior. Given that the best-performing model—containing no less than 81 predictors—attained no more than "medium" performance, the benchmarks provided by Cohen are clearly too optimistic for personality models of prosocial behavior (for similar reasoning on effects of personality, see Funder & Ozer, 2019; Gignac & Szodorai, 2016). In turn, any model explaining more than 13.9% of the variance in prosocial behavior (assuming the model is not cross-validated) is at risk of overfitting, that is, mistakenly including noise in the model, thereby modeling the data "too closely" (Rocca & Yarkoni, 2021).

This result also gives a new perspective on interpreting zero-order correlations. The strongest single predictor of SVO in the dataset (i.e., competitive jungle world view; Sibley & Duckitt, 2009) obtained a cross-validated R^2 of 9.3% (corresponding to a correlation of r = .31). By traditional rules-of-thumb (Cohen, 1988), this would constitute a "medium" effect size. Such a classification, however, is clearly misguided, as an optimal combination of 81 different traits only performs five percentage points better, corresponding to a correlation of r = .37. In this regard, and harking back to the notion of benchmarking, a single predictor explaining 9.3% can be considered an excellent predictor of prosocial behavior.

A multivariate perspective on broad and narrow traits

Personality science is populated with both broad and narrow trait constructs. Broad constructs (i.e., traits conceptualized within basic personality models, such as the Big Five and the HEXACO model; Goldberg, 1990; Ashton & Lee, 2007) seek to broadly capture the universe of potential personality traits while at the same time providing a parsimonious account of a variety of outcomes and phenomena. Narrower traits, in turn, may be tailored to specifically tap into the personality basis of a phenomenon and are thus expected to show better predictive utility (Ones & Viswesvaran, 1996). Our results showed that models fitted on all 81 traits available in our dataset clearly outperformed models fitted solely on the HEXACO dimensions. Thus, although the HEXACO model arguably provides a parsimonious, yet relatively comprehensive, representation of the personality space, it misses out on some individual variation in prosocial behavior that only more fine-grained traits seem to capture (see also Paunonen et al., 2003; Seeboth & Mõttus, 2018; Speer et al., 2022; for similar findings spanning a wide range

of outcomes). This finding differs from the meta-analysis by Thielmann et al. (2020) which found that both broad traits (e.g., the Honesty–Humility dimension of the HEXACO) and narrow traits (e.g., guilt proneness) performed similarly well as single predictors of prosocial behavior. This difference may stem from the fact that Thielmann et al. (2020) investigated univariate relations. That is, when considered in isolation, the predictive utilities of broad and narrow traits may be similar; when modelled in tandem, however, narrow traits show incrementally better predictive utility above and beyond basic personality.

Measuring to make the most of nonlinearities

Our results—and those of Oparina et al. (2022)—suggest that personality psychology benefits little from applying complex, nonlinear models. This stands in stark contrast to related fields, which have successfully applied nonlinear models to improve prediction (e.g., economics; Gogas & Papadimitriou, 2021; Nosratabadi et al., 2020). What could be holding personality psychology from reaping the same benefits of nonlinear modeling? Part of the answer may lie in the current measurement practices within personality science. When developing a novel scale, researchers are taught to strive for a symmetric distribution of responses with high variance (Clark & Watson, 1995) and to obtain large correlations with relevant criteria (Boateng et al., 2018; Devellis & Thorpe, 2021). In the context of nonlinear models, these practices have potential drawbacks. To obtain a symmetric distribution, researchers must avoid extreme items (e.g., “I often commit murder”) which almost all participants answer in the same way (i.e., “strongly disagree”). However, such extreme responses may be the locus of nonlinearity because they capture true idiosyncrasies and deviations from typical behavioral patterns. Similarly, the emphasis on establishing criterion validity through (linear) correlations will naturally prioritize traits that tend to exhibit linear relations over those with nonlinear relations. As such, the current measurement practice may prioritize measures that are likely to display linear relations.

We conjecture that two extant approaches to measurement may be more amenable to nonlinear models. First, Mõttus et al. (2017, 2020) suggest to analyze psychometric scales at the item-level instead of (or in addition to) the construct-level. Individual item responses arguably tap into specific behavioral tendencies more accurately than trait scores aggregated across multiple items. As such, they may be more sensitive to context dependencies (i.e., interactions) than aggregated trait scores. Prior work has shown this approach to be both reliable and predictive of relevant outcomes (Mõttus et al., 2017; Seeboth & Mõttus, 2018; Speer et al., 2022), albeit more demanding with regard to sample size (Revelle et al., 2021). Second, we join voices

with scholars (e.g., Bleidorn and Hopwood, 2019; Mottus et al., 2020; Rauthmann, 2020) who suggest using objective measurements (e.g., digital traces; Stachl et al., 2020) as part of personality assessment. Such measures do not undergo any selection mechanism that potentially favor linear associations and could thus be leveraged better by nonlinear models.

Limitations

Before turning to the implications, some limitations of the current investigation must be acknowledged. First, the sample size (N = 2,707), albeit large for a psychological study, is relatively small compared to many machine learning applications (which sometimes contain millions of observations; e.g., Deng et al., 2009). This may adversely affect the performance of the nonlinear models as they sometimes require a large number of observations to learn the intricacies of the data. Second, although our results suggest that model complexity is not a particularly fruitful way to improve the prediction of SVO, we cannot rule out that more complex modeling techniques (e.g., ensembles) could improve upon prediction. Indeed, there are infinite ways to fit a model, and some will undoubtedly outperform ours. We thus invite the community to beat our 13.9% benchmark.[9]

Third and finally, our conclusions rely on a single criterion, SVO, and may not generalize to other common measures of prosocial behavior. Indeed, the benchmarks we put forward should be interpreted as criterion-specific. However, the meta-analysis by Thielmann et al. (2020) revealed SVO to be strongly correlated with prosocial behavior in all economic games that capture active prosociality (see also Balliet et al., 2009). In addition, SVO has been linked to other indicators of real-life prosocial behavior, such as donating to noble causes (Van Lange et al., 2007), volunteering (Van Lange et al., 2011), and performing pro-environmental actions (Cameron et al., 1998). We can only conjecture that our conclusion about the adequacy of linear models and Pearson's correlations will generalize to other measures of prosocial behavior.[10] As for other fields of research, the generalization is even less straightforward, and we do not claim our results to hold for all kinds of outcome variables (but see Oparina et al., 2022). Rather, we hope to see researchers become aware of the possibility of investigating nonlinearities and setting benchmarks in their own fields of research.

Implications

[9] We note that there is one circumstance in which the 13.9% benchmark is impossible to break, namely if the error inherent in the SVO slider measure (i.e., the *irreducible error*) accounts for the remaining unexplained variance.

[10] Although the PPP contains measures of some economic games (e.g., trust game, dictator game), the sample sizes of these measures is much lower than that of SVO. Given that nonlinear models can require a very large number of observations, we thus decided to focus on SVO rather than economic games.

Our results can be of service to the field in multiple ways—methodologically, theoretically, and in application. First, the 13.9% benchmark can assist in model evaluation, that is, it will help researchers to answer the elusive question "how good is my model?". The benchmark also has implications for how much we emphasize personality traits in theoretical models and ascribe relevance to them in applied settings: It tells us that in a multivariate perspective, personality is substantially more predictive of behavior than is implied by single correlations (cf. Thielmann et al., 2020). In fact, the best single-trait model explained 9.3% of the variance in SVO—almost five percentage points less than the best multivariate model. Second, our findings regarding the adequacy of simple models (i.e., linear models, Pearson's correlation) support the mainstream statistical practice to consider linear effects of personality traits rather than nonlinearities and trait–trait interactions. Third, the finding that narrower traits out-predict basic personality should motivate researchers to map out more nuanced aspects of personality to complement basic personality dimensions in the prediction of prosocial behavior. In fact, our results suggest that the way toward better predictive models lies in improving our predictors (e.g., considering personality nuances; Seeboth & Mõttus, 2018) more so than in increasing the complexity of our statistical modeling practices. Importantly, however, in taking such a way forward, researchers should prevent introducing ever more construct inflation (Thielmann, Hilbig, & Zettler, 2022). Fourth, in the service of identifying relevant predictors, our contribution can inform power analysis decisions (Lakens, 2022): Because important traits tend to attain correlations of at least $r = .20$, sample size calculations should be set to detect effects of at least $r = .20$. That said, there are notable exceptions to this rule (e.g., right-wing authoritarianism; see Table 1), and the association between a trait's (univariate) correlation and (multivariate) variable importance is only moderate. We suggest using $r = .20$ only as a rough rule of thumb in the absence of better justifications.

Conclusion

Above all, we hope to have demonstrated the uses of machine learning and large-scale datasets in behavioral science (see also Bleidorn & Hopwood, 2019). As noted by Culkin (1967, p. 52), "we shape our tools, and thereafter our tools shape us." The same goes for statistical tools; machine learning can not only provide us with new answers to old questions but also point us to a myriad of exciting and novel research ventures.

Open science statement

The current investigation was preregistered prior to analysis. The preregistration is available on the Open Science Framework (https://osf.io/5qvfb/) along with the data, fitted models, and analysis scripts (https://osf.io/zcdk8/), as well as the materials and further information on the Prosocial Personality Project (https://osf.io/m2abp/).

References

Ahadi, S., & Diener, E. (1989). Multiple determinants and effect size. *Journal of Personality and Social Psychology*, *56*(3), 398–406. https://doi.org/10.1037/0022-3514.56.3.398

Aquino, K., & Reed, A. I. I. (2002). The self-importance of moral identity. *Journal of Personality and Social Psychology*, *83*(6), 1423–1440. https://doi.org/10.1037/0022-3514.83.6.1423

Ashton, M. C., & Lee, K. (2007). Empirical, theoretical, and practical advantages of the HEXACO model of personality structure. *Personality and Social Psychology Review*, *11*, 150-166. https://doi.org/10.1177/1088868306294907

Ashton, M. C., & Lee, K. (2009). The HEXACO-60: A short measure of the major dimensions of personality. *Journal of Personality Assessment*, 91(4), 340–345. https://doi.org/10.1080/00223890902935878

Ashton, M. C., Lee, K., & de Vries, R. E. (2014). The HEXACO Honesty-Humility, agreeableness, and emotionality factors: A review of research and theory. *Personality and Social Psychology Review*, *18*(2), 139–152. https://doi.org/10.1177/1088868314523838

Bainbridge, T. F., Ludeke, S. G., & Smillie, L. D. (2022). Evaluating the Big Five as an organizing framework for commonly used psychological trait scales. *Journal of Personality and Social Psychology*, *122*(4), 749–777. https://doi.org/10.1037/pspp0000395

Balliet, D., Parks, C., & Joireman, J. (2009). Social Value Orientation and Cooperation in Social Dilemmas: A Meta-Analysis. *Group Processes & Intergroup Relations*, *12*(4), 533–547. https://doi.org/10.1177/1368430209105040

Bauer, D. J., & Shanahan, M. J. (2007). Modeling complex interactions: Person-centered and variable-centered approaches. In T. D. Little, J. A. Bovaird, & N. A. Card (Eds.), *Modeling contextual effects in longitudinal studies* (pp. 255–283). Lawrence Erlbaum Associates Publishers.

Baumert, A., Beierlein, C., Schmitt, M., Kemper, C. J., Kovaleva, A., Liebig, S., & Rammstedt, B. (2014). Measuring four perspectives of justice sensitivity with two items each. *Journal of Personality Assessment*, *96*(3), 380–390. https://doi.org/10.1080/00223891.2013.836526

Baumsteiger, R., & Siegel, J. T. (2019). Measuring prosociality: The development of a Prosocial Behavioral Intentions Scale. *Journal of Personality Assessment*, *101*(3), 305–314. https://doi.org/10.1080/00223891.2017.1411918

Beierlein, C., Asbrock, F., Kauff, M., & Schmidt, P. (2014). *Die Kurzskala Autoritarismus (KSA-3): ein ökonomisches Messinstrument zur Erfassung dreier Subdimensionen*

autoritärer Einstellungen (GESIS-Working Papers, 2014/35). GESIS - Leibniz-Institut für Sozialwissenschaften. https://nbn-resolving.org/urn:nbn:de:0168-ssoar-426711

Bergstra, J., Komer, B., Eliasmith, C., Yamins, D., & Cox, D. D. (2015). Hyperopt: A Python library for model selection and hyperparameter optimization. *Computational Science & Discovery*, *8*, Article 014008. https://doi.org/10.1088/1749-4699/8/1/014008

Berkowitz, L. Daniels, L. R. (1964). Affecting the salience of the social responsibility norm: Effects of past help on the response to dependency relationships. *Journal of Abnormal and Social Psychology*, *68*, 275– 281

Berry, J. W., Worthington, E. L. J., O'Connor, L. E., Parrott, L. I. I. I., & Wade, N. G. (2005). Forgivingness, vengeful rumination, and affective traits. *Journal of Personality*, *73*(1), 183–225. https://doi.org/10.1111/j.1467-6494.2004.00308.x

Bien, J., & Tibshirani, R. (2020). *hierNet: A Lasso for Hierarchical Interactions* (1.9). https://CRAN.R-project.org/package=hierNet

Bien, J., Taylor, J., & Tibshirani, R. (2013). A lasso for hierarchical interactions. *Annals of statistics*, *41*(3), 1111–1141. https://doi.org/10.1214/13-AOS1096

Bischl, B., Binder, M., Lang, M., Pielok, T., Richter, J., Coors, S., Thomas, J., Ullmann, T., Becker, M., Boulesteix, A.-L., Deng, D., & Lindauer, M. (2023). Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges. *WIREs Data Mining and Knowledge Discovery*, *13*(2), e1484. https://doi.org/10.1002/widm.1484

Bleidorn, W., & Hopwood, C. J. (2019). Using machine learning to advance personality assessment and theory. *Personality and Social Psychology Review*, *23*(2), 190–203. https://doi.org/10.1177/1088868318772990

Boateng, G. O., Neilands, T. B., Frongillo, E. A., Melgar-Quiñonez, H. R., & Young, S. L. (2018). Best Practices for Developing and Validating Scales for Health, Social, and Behavioral Research: A Primer. *Frontiers in public health*, *6*, 149. https://doi.org/10.3389/fpubh.2018.00149

Bosco, F. A., Aguinis, H., Singh, K., Field, J. G., & Pierce, C. A. (2015). Correlational effect size benchmarks. *Journal of Applied Psychology*, *100*(2), 431–449. https://doi.org/10.1037/a0038047

Breheny, P. (2021). *ncvreg: Regularization Paths for SCAD and MCP Penalized Regression Models* (3.13.0). https://CRAN.R-project.org/package=ncvreg

Breiman, L. (1995). Better Subset Regression Using the Nonnegative Garrote. *Technometrics*, *37*(4), 373–384. https://doi.org/10.2307/1269730

591 Breiman, L. (2001). Random Forests. *Machine Learning*, *45*, 5–32.
592 https://doi.org/10.1023/A:1010933404324
593 Brunell, A. B., Davis, M. S., Schley, D. R., Eng, A. L., van Dulmen, M. H. M., Wester, K. L.,
594 & Flannery, D. J. (2013). A new measure of interpersonal exploitativeness. *Frontiers in*
595 *Psychology*, *4*, 299. https://doi.org/10.3389/fpsyg.2013.00299
596 Buss, A. H., & Perry, M. (1992). The Aggression Questionnaire. *Journal of Personality and*
597 *Social Psychology*, *63*(3), 452–459. https://doi.org/10.1037/0022-3514.63.3.452
598 Buuren, S. van, & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by
599 chained equations in R. *Journal of Statistical Software*, *45*(1), Article 1.
600 https://doi.org/10.18637/jss.v045.i03
601 Cameron, L. D., Brown, P. M., & Chapman, J. G. (1998). Social Value Orientations and deci-
602 sions to take proenvironmental Action. *Journal of Applied Social Psychology*, *28*, 675–
603 697. https://doi.org/10.1111/j.1559-1816.1998.tb01726.x
604 Campbell, W. K., Bonacci, A. M., Shelton, J., Exline, J. J., & Bushman, B. J. (2004). Psycho-
605 logical entitlement: Interpersonal consequences and validation of a self-report measure.
606 *Journal of Personality Assessment*, *83*(1), 29–45.
607 https://doi.org/10.1207/s15327752jpa8301_04
608 Caprara, G. V., Steca, P., Zelli, A., & Capanna, C. (2005). A new scale for measuring adults'
609 prosocialness. *European Journal of Psychological Assessment*, *21*(2), 77–89.
610 https://doi.org/10.1027/1015-5759.21.2.77
611 Chapman, B. P., Roberts, B., & Duberstein, P. (2011). Personality and Longevity: Knowns,
612 Unknowns, and Implications for Public Health and Personalized Medicine. *Journal of*
613 *Aging Research*, *2011*, 759170. https://doi.org/10.4061/2011/759170
614 Chen, F. F., & West, S. G. (2008). Measuring individualism and collectivism: The im-
615 portance of considering differential components, reference groups, and measurement
616 invariance. *Journal of Research in Personality*, *42*(2), 259–294.
617 https://doi.org/10.1016/j.jrp.2007.05.006
618 Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In *Proceed-*
619 *ings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and*
620 *Data Mining* (pp. 785–794). ACM. https://doi.org/10.1145/2939672.2939785
621 Chen, T., He, T., Benesty, M., Khotilovich, V., Tang, Y., Cho, H., Chen, K., Mitchell, R.,
622 Cano, I., Zhou, T., Li, M., Xie, J., Lin, M., Geng, Y., Li, Y., & Yuan, J. (2022).
623 *xgboost: Extreme Gradient Boosting* (1.6.0.1). https://CRAN.R-project.org/pack-
624 age=xgboost

Chollet, F., & others. (2015). Keras. https://keras.io

Chowdhury, R. M. M. I., & Fernando, M. (2013). The Relationships of Empathy, Moral Identity and Cynicism with Consumers' Ethical Beliefs: The Mediating Role of Moral Disengagement. *Journal of Business Ethics*, *124*(4), 677–694. https://doi.org/10.1007/s10551-013-1896-7

Christidis, A., Aelst, S. V., & Zamar, R. (2021). *nnGarrote: Non-Negative Garrote Estimation with Penalized Initial Estimators* (1.0.4). https://CRAN.R-project.org/package=nnGarrote

Clark, L. A., & Watson, D. (1995). Constructing validity: Basic issues in objective scale development. *Psychological Assessment*, *7*(3), 309–319. https://doi.org/10.1037/1040-3590.7.3.309

Coelho, G. L. H., Monteiro, R. P., Hanel, P. H. P., Vilar, R., Gouveia, V. V, & Maio, G. R. (2018). Psychometric parameters of an abbreviated vengeance scale across two countries. *Personality and Individual Differences*, *120*, 185–192. https://doi.org/10.1016/j.paid.2017.08.042

Cohen J. (1988). *Statistical Power Analysis for the Behavioral Sciences*. Routledge Academic.

Cohen, T. R., Kim, Y., & Panter, A. T. (2014). *The five-item guilt proneness scale (GP-5)*. PsyArXiv. https://doi.org/10.17605/OSF.IO/G389S

Costa, P. T., & McCrae, R. R. (1992). *Revised NEO Personality Inventory (NEO-PI–R) and NEO Five–Factor Inventory (NEO–FFI) professional manual*. Psychological Assessment Resources.

Culkin, J. M. (1967, March 18). A schoolman's guide to Marshall McLuhan. *Saturday Review*, 51–53.

Curry, O. S., Jones Chesters, M., & Van Lissa, C. J. (2019). Mapping morality with a compass: Testing the theory of 'morality-as-cooperation' with a new questionnaire. *Journal of Research in Personality*, *78*, 106–124. https://doi.org/10.1016/j.jrp.2018.10.008

Dalbert, C. (1999). The world is more just for me than generally: About the Personal Belief in a Just World Scale's validity. Social Justice Research, 12, 79-98.

Davis, M. H. (1983). Measuring individual differences in empathy: Evidence for a multidimensional approach. *Journal of Personality and Social Psychology*, *44*(1), 113–126. https://doi.org/10.1037/0022-3514.44.1.113

De Cremer, D., & Tyler, T. R. (2005). Am I respected or not? Inclusion and reputation as issues in group membership. *Social Justice Research*, *18*(2), 121–153. https://doi.org/10.1007/s11211-005-7366-3

De Matos Fernandes, C., Bakker, D., & Dijkstra, J. (2022). Assessing the test-retest reliability of the social value orientation slider measure. *Judgment and Decision Making*, *17*(1), 31–49. https://doi.org/10.1017/S1930297500009013

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition* (pp. 248–255).

Denissen, J., Soto, C., Geenen, R., John, O. P., & van Aken, M. A. G. (2021). Incorporating prosocial vs. antisocial trait content in Big Five measurement: Lessons from the Big Five Inventory-2 (BFI-2). *Journal of Research in Personality*, 104147. https://doi.org/https://doi.org/10.1016/j.jrp.2021.104147

Devellis, R. F., & Thorpe, C. T. (2021). *Scale development: Theory and Applications* (5th ed.). Sage.

DeWitt, P., & Bennett, T. (2019). *ensr: Elastic Net SearcheR* (0.1.0). https://CRAN.R-project.org/package=ensr

DeYoung, C. G., Quilty, L. C., & Peterson, J. B. (2007). Between facets and domains: 10 aspects of the Big Five. *Journal of Personality and Social Psychology*, *93*(5), 880–896. https://doi.org/10.1037/0022-3514.93.5.880

Duckitt, J., Bizumic, B., Krauss, S. W., & Heled, E. (2010). A tripartite approach to Right-Wing Authoritarianism: The Authoritarianism-Conservatism-Traditionalism model. *Political Psychology*, *31*(5), 685–715. https://doi.org/10.1111/j.1467-9221.2010.00781.x

Eisenberger, R., Lynch, P., Aselage, J., & Rohdieck, S. (2004). Who Takes the most Revenge? Individual Differences in Negative Reciprocity Norm Endorsement. *Personality and Social Psychology Bulletin*, *30*(6), 787–799. https://doi.org/10.1177/0146167204264047

Epstein, S., & O'Brien, E. J. (1985). The person–situation debate in historical and current perspective. *Psychological Bulletin*, *98*(3), 513–537. https://doi.org/10.1037/0033-2909.98.3.513

Fehr, E., & Fischbacher, U. (2002). Why social preferences matter— The impact of non-selfish motives on competition, Cooperation and Incentives. *The Economic Journal*, *112*(478), C1–C33. https://doi.org/10.1111/1468-0297.00027

690 Fisher, A., Rudin, C., & Dominici, F. (2018). All models are wrong, but many are useful:
691 Learning a variable's importance by studying an entire class of prediction models sim-
692 ultaneously. arXiv. http://arxiv.org/abs/1801.01489
693 Fleeson, W. (2004). Moving personality beyond the person-situation debate: The challenge
694 and the opportunity of within-person variability. *Current Directions in Psychological*
695 *Science, 13*(2), 83–87. https://doi.org/10.1111/j.0963-7214.2004.00280.x
696 Forsyth, D. R. (1980). A taxonomy of ethical ideologies. *Journal of Personality and Social*
697 *Psychology, 39*, 175–184.
698 Frazier, M. L., Johnson, P. D., & Fainshmidt, S. (2013). Development and validation of a
699 propensity to trust scale. *Journal of Trust Research*, *3*(2), 76–97.
700 https://doi.org/10.1080/21515581.2013.820026
701 Friedman, J., Hastie, T., Tibshirani, R., Narasimhan, B., Tay, K., Simon, N., Qian, J., &
702 Yang, J. (2021). *glmnet: Lasso and Elastic-Net Regularized Generalized Linear Models*
703 (4.1-3). https://CRAN.R-project.org/package=glmnet
704 Funder, D. C., & Ozer, D. J. (2019). Evaluating effect size in psychological research: Sense
705 and nonsense. *Advances in Methods and Practices in Psychological Science, 2*(2), 156–
706 168. https://doi.org/10.1177/2515245919847202
707 Furr, R. M., & Funder, D. C. (2021). Persons, situations, and person–situation interactions. In
708 O. P. John & R. W. Robins (Eds.), *Handbook of personality: Theory and research* (pp.
709 667–685). The Guilford Press.
710 Gerbasi, M. E., & Prentice, D. A. (2013). The Self- and Other-Interest Inventory. *Journal of*
711 *Personality and Social Psychology*, *105*(3), 495–514. https://doi.org/10.1037/a0033483
712 Gignac, G. E., & Szodorai, E. T. (2016). Effect size guidelines for individual differences re-
713 searchers. *Personality and Individual Differences, 102*, 74–78.
714 https://doi.org/10.1016/j.paid.2016.06.069
715 Gogas, P., Papadimitriou, T. (2021). Machine Learning in Economics and Finance. *Computu-*
716 *ational Economics*, *57*(1), 1–4. https://doi.org/10.1007/s10614-021-10094-w
717 Goldberg L. R. (1990). An alternative "description of personality": the big-five factor struc-
718 ture. *Journal of personality and social psychology*, *59*(6), 1216–1229.
719 https://doi.org/10.1037//0022-3514.59.6.1216
720 Goldberg, L. R. (1992). The development of markers for the Big-Five factor structure. *Psy-*
721 *chological Assessment*, *4*(1), 26–42. https://doi.org/10.1037/1040-3590.4.1.26
722 Gouveia, V. V, Vasconcelos de Oliveira, I. C., Moura Grangeiro, A. S. de, Pereira Monteiro,
723 R., & Lins de Holanda Coelho, G. (2021). The bright side of the human personality:

Evidence of a measure of prosocial traits. *Journal of Happiness Studies*, *22*(3), 1459–1480. https://doi.org/10.1007/s10902-020-00280-2

Grasmick, H. G., Tittle, C. R., Bursik, R. J., & Arneklev, B. J. (1993). Testing the core empirical implications of Gottfredson and Hirschi's general theory of crime. *Journal of Research in Crime and Delinquency*, *30*, 5–29. https://doi.org/10.1177/0022427893030001002

Helgeson, V. S. & Fritz, H. L. (1998). A theory of unmitigated communion. *Personality and Social Psychology Review*, *2*, 173–183.

Hilbig, B. E., Glöckner, A., & Zettler, I. (2014). Personality and prosocial behavior: Linking basic traits and social value orientations. *Journal of Personality and Social Psychology*, *107*(3), 529–539. https://doi.org/10.1037/a0036074

Hilbig, B. E., Thielmann, I., Zettler, I., & Moshagen, M. (2023). The Dispositional Essence of Proactive Social Preferences: The Dark Core of Personality vis-à-vis 58 Traits. *Psychological Science*, *34*(2), 201–220. https://doi.org/10.1177/09567976221116893

Ho, A. K., Sidanius, J., Kteily, N., Sheehy-Skeffington, J., Pratto, F., Henkel, K. E., Foels, R., & Stewart, A. L. (2015). The nature of social dominance orientation: Theorizing and measuring preferences for intergroup inequality using the new SDO_7 scale. *Journal of personality and social psychology*, *109*(6), 1003–1028. https://doi.org/10.1037/pspi0000033

Hoerl, A. E., & Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, *12*(1), 55-67, https://doi.org/10.1080/00401706.1970.10488634

Hogan, R. (2009). Much ado about nothing: The person-situation debate. *Journal of Research in Personality*, *43*(2), 249. https://doi.org/10.1016/j.jrp.2009.01.022

Hwang, J. Y., Plante, T., & Lackey, K. (2008). The development of the Santa Clara Brief Compassion Scale: An abbreviation of Sprecher and Fehr's Compassionate Love Scale. *Pastoral Psychology*, *56*(4), 421–428. https://doi.org/10.1007/s11089-008-0117-2

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer. https://doi.org/10.1007/978-1-4614-7138-7_1

Joireman, J., Shaffer, M. J., Balliet, D., & Strathman, A. (2012). Promotion orientation explains why future-oriented people exercise and eat healthy: Evidence from the two-factor Consideration of Future Consequences-14 Scale. *Personality and Social Psychology Bulletin*, *38*(10), 1272–1287. https://doi.org/10.1177/0146167212449362

Jones, D. N., & Paulhus, D. L. (2014). Introducing the Short Dark Triad (SD3): A brief measure of dark personality traits. *Assessment*, *21*(1), 28–41. https://doi.org/10.1177/1073191113514105

Karatzoglou, A., Smola, A., Hornik, K., N. I., Maniscalco, M. A., & Teo, C. H. (2022). *kernlab: Kernel-Based Machine Learning Lab* (0.9-31). https://CRAN.R-project.org/package=kernlab

Kaufman, S. B., Yaden, D. B., Hyde, E., & Tsukayama, E. (2019). The Light vs. Dark Triad of Personality: Contrasting Two Very Different Profiles of Human Nature. *Frontiers in psychology*, *10*, 467. https://doi.org/10.3389/fpsyg.2019.00467

Kenrick, D. T., & Funder, D. C. (1988). Profiting from controversy: Lessons from the person-situation debate. *American Psychologist*, *43*(1), 23–34. https://doi.org/10.1037/0003-066X.43.1.23

Knežević, G. (2003). Koreni amoralnosti [Roots of amoralism]. Beograd: Centar za primenjenu psihologiju, IKSI, Institut za psihologiju.

Kohn, M., & Schooler, C. (1983). *Work and Personality: An Inquiry into the Impact of Social Stratification*. Ablex.

Kuhn, M., Wing, J., Weston, S., Williams, A., Keefer, C., Engelhardt, A., Cooper, T., Mayer, Z., Kenkel, B., R Core Team, Benesty, M., Lescarbeau, R., Ziem, A., Scrucca, L., Tang, Y., Candan, C., & Hunt, T. (2022). *caret: Classification and Regression Training* (6.0-93). https://CRAN.R-project.org/package=caret

Lakens, D. (2022). Sample Size Justification. *Collabra: Psychology*, *8*(1), Article 33267. https://doi.org/10.1525/collabra.33267

Lang, M., Bischl, B., Richter, J., Schratz, P., Casalicchio, G., Coors, S., Au, Q., Binder, M., Pfisterer, F., Sonabend, R., Schneider, L., Becker, M., & Fischer, S. (2022). *mlr3: Machine Learning in R - Next Generation* (0.14.1). https://CRAN.R-project.org/package=mlr3

Lange, J., & Crusius, J. (2015). Dispositional envy revisited: Unraveling the motivational dynamics of benign and malicious envy. *Personality and Social Psychology Bulletin*, *41*, 284–294. https://doi.org/10.1177/0146167214564959

Leckelt, M., Wetzel, E., Gerlach, T. M., Ackerman, R. A., Miller, J. D., Chopik, W. J., Penke, L., Geukes, K., Küfner, A. C. P., Hutteman, R., Richter, D., Renner, K. H., Allroggen, M., Brecheen, C., Campbell, W. K., Grossmann, I., & Back, M. D. (2018). Validation of the Narcissistic Admiration and Rivalry Questionnaire Short Scale (NARQ-

S) in convenience and representative samples. *Psychological assessment*, *30*(1), 86–96. https://doi.org/10.1037/pas0000433

LeCun, Y., Bengio, Y. & Hinton, G. (2015). Deep learning. *Nature*, *521*, 436–444. https://doi.org/10.1038/nature14539

Lewin, K. (1936). *Principles of topological psychology* (F. Heider & G. M. Heider, Trans.). McGraw-Hill. https://doi.org/10.1037/10019-000

Li, R. (2023). Revisiting person–situation interactionism. Nature *Reviews Psychology*, *2*(6). https://doi.org/10.1038/s44159-022-00139-8

Li, X., Sudarsanam, N., & Frey, D. D. (2006). Regularities in data from factorial experiments. *Complexity*, *11*(5), 32–45. https://doi.org/10.1002/cplx.20123

Lim, M., & Hastie, T. (2015). Learning Interactions via Hierarchical Group-Lasso Regularization. *Journal of Computational and Graphical Statistics*, *24*(3), 627–654. https://doi.org/10.1080/10618600.2014.938812

Mandrik, C.A., & Bao, Y. (2005). Exploring the concept and measurement of general risk aversion. *Advances in Consumer Research*, *32*, 531–539.

Maples, J. L., Carter, N. T., Few, L. R., Crego, C., Gore, W. L., Samuel, D. B., Williamson, R. L., Lynam, D. R., Widiger, T. A., Markon, K. E., Krueger, R. F., & Miller, J. D. (2015). Testing whether the DSM-5 personality disorder trait model can be measured with a reduced set of items: An item response theory investigation of the Personality Inventory for DSM-5. *Psychological assessment*, *27*(4), 1195–1210. https://doi.org/10.1037/pas0000120

Marcus, D. K., Zeigler-Hill, V., Mercer, S. H., & Norris, A. L. (2014). The psychology of spite and the measurement of spitefulness. *Psychological Assessment*, *26*(2), 563–574. https://doi.org/10.1037/a0036039

McClintock, C. G., & Liebrand, W. B. (1988). Role of interdependence structure, individual value orientation, and another's strategy in social decision making: A transformational analysis. *Journal of Personality and Social Psychology*, *55*(3), 396–409. https://doi.org/10.1037/0022-3514.55.3.396

McCullough, M. E., Emmons, R. A., & Tsang, J.-A. (2002). The grateful disposition: A conceptual and empirical topography. *Journal of Personality and Social Psychology*, *82*(1), 112–127. https://doi.org/10.1037//0022-3514.82.1.112

Meinshausen, N. (2007). Relaxed Lasso. *Computational Statistics & Data Analysis*, *52*(1), 374-393. https://doi.org/10.1016/j.csda.2006.12.019

Miller, A. (2002). *Subset Selection in Regression* (2nd ed.). Chapman and Hall. https://doi.org/10.1201/9781420035933

Miller, T. L. (2020). *leaps: Regression Subset Selection* (3.1). https://CRAN.R-project.org/package=leaps

Mischel, W. (1968). *Personality and assessment.* John Wiley & Sons Inc.

Molnar, C., & Schratz, P. (2022). *iml: Interpretable Machine Learning* (0.11.1). https://CRAN.R-project.org/package=iml

Moore, C., Detert, J. R., Treviño, L. K., Baker, V. L., & Mayer, D. M. (2012). Why employees do bad things: Moral disengagement and unethical organizational behavior. *Personnel Psychology*, *65*, 1–48. https://doi.org/10.1111/j.1744-6570.2011.01237.x

Moshagen, M., Zettler, I., & Hilbig, B. E. (2020). Measuring the dark core of personality. *Psychological Assessment*, *32*, 182-196. https://doi.org/10.1037/pas0000778

Mõttus, R., Sinick, J., Terracciano, A., Hřebíčková, M., Kandler, C., Ando, J., Mortensen, E. L., Colodro-Conde, L., & Jang, K. L. (2019). Personality characteristics below facets: A replication and meta-analysis of cross-rater agreement, rank-order stability, heritability, and utility of personality nuances. *Journal of Personality and Social Psychology, 117*(4), e35–e50. https://doi.org/10.1037/pspp0000202

Mõttus, R., Wood, D., Condon, D. M., Back, M. D., Baumert, A., Costantini, G., Epskamp, S., Greiff, S., Johnson, W., Lukaszewski, A., Murray, A., Revelle, W., Wright, A. G. C., Yarkoni, T., Ziegler, M., & Zimmermann, J. (2020). Descriptive, predictive and explanatory personality research: Different goals, different approaches, but a shared need to move beyond the big few traits. *European Journal of Personality, 34*(6), 1175–1201. https://doi.org/10.1002/per.2311

Muggeo, V. M. R. (2023). *segmented: Regression Models with Break-Points / Change-Points (with Possibly Random Effects) Estimation* (1.6-4). https://cran.r-project.org/web/packages/segmented/index.html

Murphy, R. O., & Ackermann, K. A. (2014). Social value orientation: Theoretical and measurement issues in the study of social preferences. *Personality and Social Psychology Review*, *18*(1), 13–41. https://doi.org/10.1177/1088868313501745

Murphy, R. O., Ackermann, K. A., & Handgraaf, M. J. J. (2011). Measuring social value orientation. *Judgment and Decision Making*, *6*(8), 771–781.

Nosratabadi, S., Mosavi, A., Duan, P., Ghamisi, P., Filip, F., Band, S. S., Reuter, U., Gama, J., & Gandomi, A. H. (2020). Data Science in Economics: Comprehensive Review of

Advanced Machine Learning and Deep Learning Methods. *Mathematics*, *8*(10), 1799. https://doi.org/10.3390/math8101799

O'Meara, A., Davies, J., & Hammond, S. (2011). The psychometric properties and utility of the Short Sadistic Impulse Scale (SSIS). *Psychological Assessment, 23*(2), 523–531. https://doi.org/10.1037/a0022400

Ones, D. S., & Viswesvaran, C. (1996). Bandwidth-fidelity dilemma in personality measurement for personnel selection. *Journal of Organizational Behavior, 17,* 609–626. https://doi.org/10.1002/(SICI)1099-1379(199611)17:6_609::AID-JOB1828_3.0.CO;2-K

Oparina, E., Kaiser, C. Gentile, N., Tkatchenko, A., Clark, A. E., De Neve, J.-E., & D'Ambrosio, C. (2022). *Human Wellbeing and Machine Learning*. arXiv. https://doi.org/10.48550/arxiv.2206.00574

Pandey, S., & Elliott, W. (2010). Suppressor variables in social work research: Ways to identify in multiple regression models. *Journal of the Society for Social Work and Research, 1*(1), 28–40. https://doi.org/10.5243/jsswr.2010.2

Paunonen, S. V., Haddock, G., Forsterling, F., & Keinonen, M. (2003). Broad versus narrow personality measures and the prediction of behaviour across cultures. European Journal of Personality, 17(6), 413–433. https://doi.org/10.1002/per.496

Perugini, M., Gallucci, M., Presaghi, F., & Ercolani, A. P. (2003). The personal norm of reciprocity. *European Journal of Personality*, *17*(4), 251–283. https://doi.org/10.1002/per.474

Pervin, L. A. (1989). Persons, Situations, Interactions: The History of a Controversy and a Discussion of Theoretical Models. *The Academy of Management Review*, *14*(3), 350–360. https://doi.org/10.2307/258172

Peterson, C., Park, N., & Seligman, M.E. P. (2005). Assessment of character strengths. In G. P. Koocher, J. C. Norcross, & S. S. Hill III (Eds.), *Psychologists' desk reference* (2nd ed., pp. 93–98). Oxford University Press.

Probst, P., Boulesteix, A-L., & Bischl, B. (2019). Tunability: Importance of Hyperparameters of Machine Learning Algorithms. *Journal of Machine Learning Research*, *20*(53), 1–32. http://jmlr.org/papers/v20/18-444.html

R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

888 Raine, A., & Uh, S. (2019). The Selfishness Questionnaire: Egocentric, adaptive, and patho-
889 logical forms of selfishness. *Journal of Personality Assessment*, *101*(5), 503–514.
890 https://doi.org/10.1080/00223891.2018.1455692
891 Rauthmann, J. F. (2020). A (more) behavioral science of personality in the age of multi-
892 modal sensing, big data, machine learning, and artificial intelligence. *European Journal*
893 *of Personality*, *34*(5), 593–598. https://doi.org/10.1002/per.2310
894 Revelle, W., Dworak, E. M., & Condon, D. M. (2021). Exploring the persome: The power of
895 the item in understanding personality structure. *Personality and Individual Differences*,
896 *169*, Article 109905. https://doi.org/10.1016/j.paid.2020.109905
897 Richard, F. D., Bond, C. F., Jr., & Stokes-Zoota, J. J. (2003). One hundred years of social
898 psychology quantitatively described. *Review of General Psychology*, *7*, 331–363.
899 https://doi.org/10.1037/1089-2680.7.4.331
900 Rocca, R., & Yarkoni, T. (2021). Putting psychology to the test: Rethinking model evaluation
901 through benchmarking and prediction. *Advances in Methods and Practices in Psycho-*
902 *logical Science*, *4*(3), 25152459211026864.
903 https://doi.org/10.1177/25152459211026864
904 Rosenbusch, H., Soldner, F., Evans, A.M. and Zeelenberg, M. (2021), Supervised machine
905 learning methods in psychology: A practical introduction with annotated R code. Soc
906 Personal Psychol Compass, 15: e12579. https://doi.org/10.1111/spc3.12579
907 Rothmund, T., Bromme, L., & Azevedo, F. (2020). Justice for the people? How justice sensi-
908 tivity can foster and impair support for populist radical-right parties and politicians in
909 the United States and in Germany. *Political Psychology*, *41*(3), 479–497.
910 https://doi.org/10.1111/pops.12632
911 RStudio Team. (2020). *RStudio: Integrated Development for R*. RStudio, PBC.
912 http://www.rstudio.com/
913 Schmitt, M. J., Neumann, R., & Montada, L. (1995). Dispositional Sensitivity to Befallen In-
914 justice. *Social Justice Research*, *8*(4), 385–407. https://doi.org/10.1007/bf02334713
915 Schmitt, M., Baumert, A., Gollwitzer, M., & Maes, J. (2010). The Justice Sensitivity Inven-
916 tory: Factorial validity, location in the personality facet space, demographic pattern,
917 and normative data. *Social Justice Research*, *23*(2–3), 211–238.
918 https://doi.org/10.1007/s11211-010-0115-2
919 Schmitt, M., Maes, J., & Schmal, A. (1995). *Gerechtigkeit als innerdeutsches Problem: Ein-*
920 *stellungen zu Verteilungsprinzipien, Ungerechtigkeitssensibilität und Glaube an eine*

gerechte Welt als Kovariate. PsychArchives. https://doi.org/10.23668/psycharchives.8917

Seeboth, A., & Mõttus, R. (2018). Successful explanations start with accurate descriptions: Questionnaire items as personality markers for more accurate predictions. *European Journal of Personality, 32*(3), 186–201. https://doi.org/10.1002/per.2147

Seuntjens, T. G., Zeelenberg, M., van de Ven, N., & Breugelmans, S. M. (2015). Dispositional greed. *Journal of Personality and Social Psychology*, *108*(6), 917–933. https://doi.org/10.1037/pspp0000031

Shweder, R. A. (1975). How relevant is an individual difference theory of personality? Journal of Personality, 43, 455-484

Sibley, C. G., & Duckitt, J. (2009). Big-five personality, social worldviews, and ideological attitudes: further tests of a dual process cognitive-motivational model. *The Journal of social psychology, 149*(5), 545–561. https://doi.org/10.1080/00224540903232308

Singelis, T. M., Triandis, H. C., Bhawuk, D. P. S., & Gelfand, M. J. (1995). Horizontal and vertical dimensions of individualism and collectivism: A theoretical and measurement refinement. *Cross-Cultural Research*, *29*(3), 240–275. https://doi.org/10.1177/106939719502900302

Smola, A.J., Schölkopf, B. (2004). A tutorial on support vector regression. *Statistics and Computing 14*, 199–222. https://doi.org/10.1023/B:STCO.0000035301.49549.88

Soto, C. J. & John, O. P. (2017). The next Big Five Inventory (BFI-2): Developing and assessing a hierarchical model with 15 facets to enhance bandwidth, fidelity, and predictive power. *Journal of Personality and Social Psychology*, *113*, 117–143. https://doi.org/10.1037/pspp0000096

Speer, A. B., Christiansen, N. D., Robie, C., & Jacobs, R. R. (2022). Measurement specificity with modern methods: Using dimensions, facets, and items from personality assessments to predict performance. *Journal of Applied Psychology, 107*(8), 1428–1439. https://doi.org/10.1037/apl0000618

Spinella M. (2007). Normative data and a short form of the Barratt Impulsiveness Scale. *The International journal of neuroscience, 117*(3), 359–368. https://doi.org/10.1080/00207450600588881

Stachl, C., Au, Q., Schoedel, R., Gosling, S. D., Harari, G. M., Buschek, D., Völkel, S. T., Schuwerk, T., Oldemeier, M., Ullmann, T., & Hussmann, H. (2020). Predicting personality from patterns of behavior collected with smartphones. *Proceedings of the National Academy of Sciences, 117*(30), 17680–17687. https://doi.org/10.1073/pnas.1920484117

Steegen, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*, *11*(5), 702–712. https://doi.org/10.1177/1745691616658637

Tangney, J. P., Baumeister, R. F., & Boone, A. L. (2004). High self-control predicts good adjustment, less pathology, better grades, and interpersonal success. *Journal of Personality*, *72*(2), 271–322. https://doi.org/10.1111/j.0022-3506.2004.00263.x

Thielmann, I., & Pfattheicher, S. (2022). Editorial overview: Current and new directions in the study of prosociality. *Current opinion in psychology*, *47*, 101355. https://doi.org/10.1016/j.copsyc.2022.101355

Thielmann, I., Hilbig, B. E., & Zettler, I. (2022). The dispositional basis of human prosociality. *Current opinion in psychology*, *43*, 289–294. https://doi.org/10.1016/j.copsyc.2021.08.009

Thielmann, I., Moshagen, M., Hilbig, Benjamin, E., & Zettler, I. (2022). On the Comparability of Basic Personality Models: Meta-Analytic Correspondence, Scope, and Orthogonality of the Big Five and HEXACO Dimensions. *European Journal of Personality*, *36*(6), 870–900. https://doi.org/10.1177/08902070211026793

Thielmann, I., Spadaro, G., & Balliet, D. (2020). Personality and prosocial behavior: A theoretical framework and meta-analysis. *Psychological Bulletin*, *146*(1), 30–90. https://doi.org/10.1037/bul0000217

Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society. Series B (Methodological)*, *58*(1), 267–288. http://www.jstor.org/stable/2346178

Turner, R. H. (1988). Personality in Society: Social Psychology's Contribution to Sociology. *Social Psychology Quarterly*, *51*(1), 1–10. https://doi.org/10.2307/2786979

Van Lange, P. A. M., Bekkers, R., Schuyt, T. N. M., & Van Vugt, M. (2007). From Games to Giving: Social Value Orientation Predicts Donations to Noble Causes. *Basic and Applied Social Psychology*, *29*(4), 375–384. https://doi.org/10.1080/01973530701665223

Van Lange, P. A. M., Schippers, M., & Balliet, D. (2011). Who volunteers in psychology experiments? An empirical review of prosocial motivation in volunteering. *Personality and Individual Differences*, *51*, 279–284. https://doi.org/10.1016/j.paid.2010.05.038

Van Rossum, G., & Drake, F. L. (2009). *Python 3 Reference Manual*. CreateSpace.

Vize, C. E., Sharpe, B. M., Miller, J. D., Lynam, D. R., & Soto, C. J. (2022). Do the Big Five personality traits interact to predict life outcomes? Systematically testing the

prevalence, nature, and effect size of trait-by-trait moderation. *European Journal of Personality*, *0*(0). https://doi.org/10.1177/08902070221111857

Weigel, R. H., Hessing, D. J., & Elffers, H. (1999). Egoism: Concept, measurement and implications for deviance. *Psychology, Crime & Law*, *5*(4), 349–378. https://doi.org/10.1080/10683169908401777

Wright, M. N., Wager, S., & Probst, P. (2022). *ranger: A Fast Implementation of Random Forests* (0.14.1). https://CRAN.R-project.org/package=ranger

Yamagishi, T., & Yamagishi, M. (1994). Trust and commitment in the United States and Japan. *Motivation and Emotion*, *18*(2), 129–166. https://doi.org/10.1007/BF02249397

Yarkoni, T., & Westfall, J. (2017). Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. Perspectives on Psychological Science, 12(6), 1100–1122. https://doi.org/10.1177/1745691617693393

Zhang, C.-H. (2010). Nearly unbiased variable selection under minimax concave penalty. *The Annals of Statistics*, *38*(2), 894-942. https://doi.org/10.1214/09-AOS729

Zou, H. (2006). The Adaptive Lasso and Its Oracle Properties. *Journal of the American Statistical Association*, *101*(476), 1418-1429. https://doi.org/10.1198/016214506000000735

Zou, H., & Hastie, T. (2005). Regularization and Variable Selection via the Elastic Net. *Journal of the Royal Statistical Society. Series B (Statistical Methodology)*, *67*(2), 301–320. http://www.jstor.org/stable/3647580

1008 **Table 1.** Personality predictors included in the analysis.

Construct	Scale	Items	*M* (*SD*)	*r*	*VI*
Basic traits					
Dark factor of personality (D)	D-70 (Moshagen et al., 2020) Example: *I believe that lying is necessary to maintain a competitive advantage over others.*	70	2.16 (0.45)	.090	1.013
HEXACO honesty–humility	HEXACO-60 (Ashton & Lee, 2009) Example: *I think that I am entitled to more respect than the average person is.* (R)	10	3.63 (0.63)	.063	1.002
HEXACO emotionality	HEXACO-60 (Ashton & Lee, 2009) Example: *I worry a lot less than most people do.* (R)	10	3.17 (0.61)	-.003	1.001
HEXACO extraversion	HEXACO-60 (Ashton & Lee, 2009) Example: *The first thing that I always do in a new place is to make friends.*	10	3.25 (0.61)	.004	1.000
HEXACO agreeableness	HEXACO-60 (Ashton & Lee, 2009) Example: *People think of me as someone who has a quick temper.* (R)	10	3.23 (0.53)	.012	1.000
HEXACO conscientiousness	HEXACO-60 (Ashton & Lee, 2009) Example: *I do only the minimum amount of work needed to get by.* (R)	10	3.70 (0.51)	.000	1.001
HEXACO openness to experience	HEXACO-60 (Ashton & Lee, 2009) Example: *People have often told me that I have a good imagination.*	10	3.25 (0.68)	.015	1.001
FFM agreeableness (NEO-FFI)	NEO Five Factor Inventory (Costa & McCrae, 1992) Example: *I try to be courteous to everyone I meet.*	12	3.75 (0.50)	.033	1.001
FFM agreeableness (BFAS)	Big Five Aspects Scales (DeYoung et al., 2007) Example: *I am not interested in other people's problems.* (R)	20	3.89 (0.45)	.038	1.000
FFM agreeableness (BFI)	Big Five Inventory-2 (BFI-2; Soto & John, 2017) Example: *I am someone who has a forgiving nature.*	12	3.69 (0.53)	.028	1.000
FFM agreeableness (IPIP)	International Personality Item Pool (IPIP-50; Goldberg, 1992) Example: *I make people feel at ease.*	10	3.89 (0.57)	.028	1.000
BFI Honesty–Humility	Big Five Inventory-2 Agreeableness: Honesty–Humility facet (Denissen et al., 2021) Example: *I am someone who is humble, doesn't think they're better than others.*	4	4.30 (0.51)	.016	1.000
PID-5 antagonism	PID-5 (Maples et al., 2015) Example: *I'll stretch the truth if it's to my advantage.*	20	1.52 (0.40)	.039	1.000
PID-5 detachment	PID-5 (Maples et al., 2015) Example: *I don't like spending time with others.*	20	1.66 (0.48)	.008	1.000

PID-5 disinhibition	PID-5 (Maples et al., 2015) Example: *I make promises that I don't really intend to keep.*	20	1.73 (0.38)	.003	1.001
PID-5 negative affect	PID-5 (Maples et al., 2015) Example: *I get emotional over every little thing*.	20	1.88 (0.43)	.004	0.999
PID-5 psychoticism	PID-5 (Maples et al., 2015) Example: *I often have thoughts that make sense to me but that other people say are strange.*	20	1.53 (0.47)	-.001	1.000
Narrow traits					
Aggressiveness	Buss-Perry Aggression Questionnaire (Buss & Perry, 1992) Example: *At times I feel I have gotten a raw deal out of life.*	29	2.28 (0.50)	.024	0.999
Altruism	Prosocialness Scale (Caprara et al., 2005) Example: *I try to help others.*	16	3.66 (0.59)	.039	1.003
Amoralism-crudelia	AMR40 (Knežević, 2003) Example: *Doing good deeds brings joy to the heart.* (R)	13	1.96 (0.53)	.058	1.006
Amoralism-frustralia	AMR40 (Knežević, 2003) Example: *A smart person trusts no one.*	14	2.44 (0.49)	.055	1.001
Belief in a just world	Global and Personal Belief in a Just World Scale (Dalbert, 1999) Example: *I believe that, by and large, people get what they deserve*.	13	3.11 (0.64)	.008	0.999
Beliefs in reciprocity	Personal Norm of Reciprocity Scale (Perugini et al., 2003) Example: *If I work hard, I expect it will be repaid.*	9	3.18 (0.47)	-.003	1.000
Collectivism	Individualism and Collectivism Scale-Revised (Singelis et al., 1995) Example: *My happiness depends very much on the happiness of those around me*.	16	3.34 (0.49)	.009	1.000
Compassion	Santa Clara Brief Compassion Scale (Hwang et al., 2008) Example: *I tend to feel compassion for people, even though I do not know them*.	5	3.22 (0.76)	.032	1.001
Competitive jungle world view	Two World View scale, short form (Sibley & Duckitt, 2009) Example: *It's a dog-eat-dog world where you have to be ruthless at times*	6	2.12 (0.59)	.093	1.015
Competitiveness	Individualism–Collectivism Scale (Chen & West, 2008) Example: *I feel that I have to be better than everyone else.*	8	2.66 (0.75)	.027	1.000
Concern for reputation	Concern for Reputation Scale (de Cremer & Tyler, 2005) Example: *I wish to have a good reputation.*	7	3.21 (0.78)	.003	1.001

Consideration of future consequences	Consideration of Future Consequences Scale–14 (Joireman et al., 2012) Example: *When I make a decision, I think about how it might affect me in the future.*	12	3.54 (0.52)	.014	1.002
Dangerous world view	Two World View scale, short form (Sibley & Duckitt, 2009) Example: *There are many dangerous people in our society who will attack someone out of pure meanness, for no reason at all*	6	2.89 (0.78)	.023	1.001
Egoism	Egoism Scale (Weigel et al., 1999) Example: *It is hard to get ahead without cutting corners here and there.*	12	2.47 (0.61)	.047	1.000
Empathy	Interpersonal Reactivity Index (Davis, 1983) Example: *I often have tender, concerned feelings for people less fortunate than me.*	14	3.65 (0.52)	.042	1.001
Envy	Benign and Malicious Envy Scale: Malicious Envy subscale (Lange & Crusius, 2015) Example: *If other people have something that I want for myself, I wish to take it away from them.*	5	1.79 (0.70)	.027	1.000
Exploitativeness	Interpersonal Exploitativeness Scale (Brunell et al., 2013) Example: *It doesn't bother me to benefit at someone else's expense.*	6	1.78 (0.69)	.077	1.003
Fairness concerns	Values in Action Inventory of Strengths: Fairness subscale (Peterson et al., 2005) Example: *I treat all people equally regardless of who they might be.*	5	3.99 (0.65)	.033	1.000
Faith in humanity	Light Triad Scale (Kaufman et al., 2019) Example: *I think people are mostly good.*	4	3.17 (0.78)	.028	1.000
Forgiveness	Trait Forgivingness Scale (Berry et al., 2005) Example: *People close to me probably think I hold a grudge too long*. (R)	10	3.29 (0.60)	.034	1.000
Gratitude	Gratitude Questionnaire–6 (McCullough et al., 2001) Example: *I have so much in life to be thankful for.*	6	3.86 (0.63)	.027	1.000
Greed	Dispositional Greed Scale (Seuntjens et al., 2015) Example: *I always want more.*	7	2.38 (0.78)	.055	1.001
Guilt proneness	Five-Item Guilt Proneness Scale (GP-5; Cohen et al., 2014) Example: *You secretly commit a felony. What is the likelihood you would feel remorse about breaking the law?*	5	3.97 (0.84)	.049	1.001
Humanism	Light Triad Scale (Kaufman et al., 2019) Example: *I enjoy listening to people from all walks of life.*	4	3.45 (0.60)	.024	1.000
Humility	Values in Action Inventory of Strengths: Humility subscale (Peterson & Seligman, 2004) Example: *I do not act as if I am a special person.*	5	3.45 (0.65)	.004	1.000

Impulsivity	Barrat Impulsiveness Scale (BIS-15; Spinella, 2007) Example: *I do things without thinking.*	15	2.41 (0.51)	-.001	1.000
Justice sensitivity – other	Justice Sensitivity Scale, short form (Baumert et al., 2014) Example: *I feel guilty when I enrich myself at the cost of others.*	6	3.37 (0.67)	.046	1.012
Kantianism	Light Triad Scale (Kaufman et al., 2019) Example: *I would like to be authentic even if it may damage my reputation.*	4	3.91 (0.61)	.018	1.000
Machiavellianism	Short Dark Triad (SD3; Jones & Paulhus, 2014) Example: *Whatever it takes, you must get the important people on your side.*	9	2.93 (0.66)	.062	1.001
Moral disengagement	Propensity to Morally Disengage Scale (Moore et al., 2012) Example: *It is okay to spread rumors to defend those you care about.*	8	1.95 (0.59)	.018	0.999
Moral idealism	Ethics Position Questionnaire: Moral ideology subscale (Forsyth, 1980) Example: *If an action could harm an innocent other, then it should not be done.*	6	4.23 (0.62)	.023	0.999
Moral identity	Self-Importance of Moral Identity Scale (Aquino & Reed, 2002) Example: *It would make me feel good to be a person who has these characteristics [i.e., caring, compassionate, fair, friendly - generous, helpful, hardworking, honest, kind].*	10	3.39 (0.55)	.012	0.999
Moral relativism	Ethics Position Questionnaire (Forsyth, 1980) Example: *Questions of what is ethical for everyone can never be resolved since what is moral or immoral is up to the individual.*	6	3.21 (0.65)	-.006	1.000
Morality	Morality-as-Cooperation Questionnaire: Judgment subscale (Curry et al., 2019) Example: *You should always be loyal to your family.*	21	3.60 (0.39)	.007	1.001
Narcissism (SD3)	Short Dark Triad (SD3; Jones & Paulhus, 2014) Example: *Many group activities tend to be dull without me.*	9	2.45 (0.58)	.017	1.000
Narcissism (NARQ)	Narcissistic Admiration and Rivalry Questionnaire, short form (NARQ-S; Leckelt et al., 2018) Example: *I deserve to be seen as a great personality.*	6	2.13 (0.72)	.054	1.002
Negative reciprocity	Personal Norm of Reciprocity Scale (Perugini et al., 2003) Example: *If I suffer a serious wrong, I will take my revenge as soon as possible, no matter what the costs*	9	2.58 (0.76)	.043	1.001

Negative reciprocity norm endorsement	Negative reciprocity norm endorsement scale (Eisenberger et al., 2004) Example: *If someone dislikes you, you should dislike them.*	14	2.47 (0.72)	.056	1.006
Normlessness	Normlessness Scale (Kohn & Schooler, 1983) Example: *It's alright to do anything you want, as long you stay out of trouble.*	4	2.38 (0.63)	.036	1.000
Positive reciprocity	Personal Norm of Reciprocity Scale (Perugini et al., 2003) Example: *If someone does a favour for me, I am ready to return it*	9	4.07 (0.47)	.028	1.003
Prosocial behavior intentions	Prosocial Behavior Intentions Scale (Baumsteiger & Siegel, 2019) Example: *Comfort someone I know after they experience a hardship*	4	6.16 (0.87)	.000	1.000
Prosocial personality	Prosocial Personality Inventory (Gouveia et al., 2021) Example: *I forgive people easily.*	18	3.59 (0.41)	.025	1.000
Psychological entitlement	Psychological Entitlement Scale (Campbell et al., 2004) Example: *People like me deserve an extra break now and then.*	9	2.70 (0.67)	.040	1.003
Psychopathy	Short Dark Triad (SD3; Jones & Paulhus, 2014) Example: *Payback needs to be quick and nasty.*	9	2.06 (0.59)	.031	1.000
Right-wing authoritarianism (ACT)	Authoritarianism-Conservatism-Traditionalism Scales, short form: Conservatism scale (ACT; Duckitt et al., 2010) Example: *It's great that many young people today are prepared to defy authority.*	6	2.49 (0.74)	.022	1.000
Right-wing authoritarianism (KSA)	Kurzskala Autoritarismus (KSA-3; Beierlein et al., 2014) Example: *Rules in society should be enforced without pity.*	9	2.92 (0.76)	.035	1.005
Risk propensity	General Risk Aversion Scale (Mandrik & Bao, 2005) Example: *Before I make a decision, I like to be absolutely sure how things will turn out.*	6	3.49 (0.63)	.001	1.000
Sadism	Short Sadistic Impulse Scale (SSIS; O'Meara et al., 2011) Example: *Hurting people would be exciting*	10	1.47 (0.50)	.022	1.000
Self-centeredness	Self-Control Scale: Self-centeredness subscale (Grasmick et al., 1993) Example: *If things I do upset people, it's their problem not mine.*	4	2.34 (0.76)	.024	1.000
Self-control	Brief Self-Control Scale (Tangney et al., 2004) Example: *I often act without thinking through all the alternatives.* (R)	13	3.26 (0.53)	-.005	1.000
Self-interest	Self-Other Interest Inventory (Gerbasi & Prentice, 2013)	9	3.28 (0.64)	.032	1.001

	Example: *I am constantly looking for ways to get ahead.*				
Selfishness	Selfishness Questionnaire (Raine & Uh, 2019) Example: *When it comes to helping myself or helping others, I tend to help myself.*	24	2.46 (0.64)	.084	1.009
Sensitivity to befallen injustice	Sensitivity to befallen injustice scale (Schmitt et al., 1995) Example: *I am taken advantage of by others.*	18	2.51 (0.76)	.015	0.999
Social dominance orientation	SDO_7 Scale (Ho et al., 2015) Example: *An ideal society requires some groups to be on top and others to be on the bottom.*	8	2.27 (0.67)	.042	1.001
Social responsibility	Social Responsibility Scale (Berkowitz & Daniels, 1964) Example: *I am the kind of person that people can count on.*	8	4.21 (0.46)	.001	0.999
Social welfare concerns	Theories of Self-Relative-to-Other Behavior Scale: Prosocial subscale (Gerbasi & Prentice, 2013) Example: *It is the total amount of benefit that everyone receives that matters most.*	3	3.45 (0.68)	.025	1.000
Spitefulness	Spitefulness Scale (Marcus et al., 2014) Example: *It is sometimes worth a little suffering on my part to see others receive the punishment they deserve.*	17	1.79 (0.52)	.030	1.000
Trait cynicism	Cynicism Scale (Chowdhury & Fernando, 2013) Example: *Most people would tell a lie if they could gain by it.*	5	3.39 (0.73)	.026	1.000
Trust propensity (GTS)	General Trust Scale (GTS; Yamagishi & Yamagishi, 1994) Example: *Most people are trustworthy.*	6	3.47 (0.66)	.024	1.000
Trust propensity (TPS)	Trust Propensity Scale (TPS; Frazier et al., 2013) Example: *I usually trust people until they give me a reason not to trust them.*	4	3.40 (0.76)	.013	1.000
Ultimate justice beliefs	Ultimate justice beliefs scale (Schmitt et al., 1995) Example: *Someday, everyone has to make amends for the harm they have done*	6	3.43 (0.87)	.003	1.000
Unmitigated communion	Revised Unmitigated Communion Scale (Helgeson & Fritz, 1998) Example: *I always place the needs of others above my own.*	9	3.32 (0.60)	.026	1.001
Vengefulness	Vengeance Scale, short (Coelho et al., 2018) Example: *I try to even the score with anyone who hurts me.*	10	2.15 (0.76)	.042	1.000
Victim sensitivity	Justice Sensitivity Scale (Schmitt et al., 2010) Example: *It bothers me when others receive something that ought to be mine.*	10	3.28 (0.75)	.023	1.001

Note. Correlations (r) are computed as the square root of cross-validated R^2 values, pooled across imputed datasets. Variable importance values are derived from the random forest model using permutation and pooled across imputed datasets. The variable importances from

1012 the remaining models are available on the OSF (https://osf.io/zcdk8/). VI = variable im-
1013 portance. (R) = reverse-keyed. FFM = Five Factor Model. PID-5 = Personality Inventory for
1014 DSM-5.

Table 2. Overview of statistical models and their hyperparameters.

Statistical model	Basic description	Hyperparameters
Linear		
Forward-backward selection	Finds the optimal set of predictors by iteratively adding and removing a predictor. (Miller, 2002).	nvmax
Lasso	Shrinks the coefficients of (or removes) less important predictors by minimizing both the error and $\sum\|\beta\|$ (Tibshirani, 1996).	lambda
Ridge	Shrinks the coefficients of less important predictors by minimizing both the error and $\sum\beta^2$ (Hoerl & Kennard, 1970).	lambda
Elastic net	Uses a weighted combination of the lasso and ridge regression (Zou & Hastie, 2005).	lambda, alpha
Relaxed lasso	Applies the lasso to select a subset of predictors and then computes the OLS coefficients for these predictors (Meinshausen, 2007).	lambda, gamma
Adaptive lasso	Applies the lasso but weights the penalty on each predictor by $1/\left\|\hat{\beta}_{\mathrm{OLS}}\right\|$ (Zou, 2006).	lambda
Smoothly clipped absolute deviation (SCAD)	Applies the lasso but uses a penalty that prioritizes small predictor sets while allowing for large coefficients (Fan & Li, 2001).	lambda
Minimax concave penalty (MCP)	Applies the lasso but uses a penalty that prioritizes small predictor sets while allowing for large coefficients (Zhang, 2010).	lambda
Non-negative garrote	Estimates the OLS coefficients and subsequently shrinks the coefficients of less important predictors (Breiman, 1995).	lambda
Linear with pairwise interactions		
glinternet	Lasso regression that also considers interactions between predictors (Lim & Hastie, 2015)	lambda
hierNet	Lasso regression that also considers interactions between predictors (Bien et al., 2013)	lambda
Nonlinear		
XGBoost	First fits a series of decision trees, each tree made to improve upon the errors of its successor, then averages their individual predictions (Chen & Guestrin, 2016).	eta, nrounds, max_depth, min_child_weight, subsample, colsample_bytree, gamma, lambda, alpha
Random forest	Fits a set of decision trees on bootstrap samples of the data and averages their individual predictions (Breiman, 2001).	num.trees, mtry.ratio, min.node.size, max.depth, sample.fraction, replace
Deep neural network	Connects the predictors to the outcome via a network of nodes whose connections encode abstract patterns in the data (LeCun et al., 2015).	layers, nodes, dropout, batch_size, learning_rate
Support vector regression (SVR)	First transforms predictors into a larger non-dimensional space and subsequently attempts to find a function which is both flat and describes all data points within a given margin of error (Smola & Schölkopf, 2004).	sigma, C, epsilon, kernel, degree

Note. Some models include additional hyperparameters which we do not consider. The hyperparameters are named according to the software used to implement the models (see section "Software"). All linear models within interactions obey the hierarchy constraint, that is, a pairwise interaction could only be included in the model if both main effects were also included. (Bien et al., 2013). The adaptive lasso was fitted using weights from OLS regression. OLS = ordinary least squares.